Surfaces, Shapes, Spins and Orbits: Archival Tests for Artifacts Among the Small Bodies

Charles C. Norton

Abstract

An artifact among the asteroids, comets and meteoroids would most likely be found in the archives that already hold its brightness, colour, shape, spin and orbit. We apply thirteen tests to them. Of 315 radar detections, every albedo above 0.3 belongs to a metal-rich or enstatite body and none is the specular return of a flat reflector. Gaia DR3 reflectance spectra of 58208 bodies hold no solar-cell surface, which injections recover at 10 per cent of a body's light in 88 per cent of cases. In 4723 ALCDEF sessions no brightening recurs at a body's rotation period beyond chance. No ATLAS phase curve is flat at a high albedo, and no NEOWISE fit stays hotter than sunlight allows at every epoch. Reduced to zero phase, no lightcurve amplitude among 31008 records exceeds the 2.5 mag of a 10 : 1 plate. The longest straight limbs in 395 stellar occultations lie on bodies whose shape models carry flat facets. No kilometre body spins faster than ten minutes or needs more than 100 kPa to hold together. The one triple of asteroids that share a period, magnitude and amplitude on unrelated orbits is three S-types at the spin barrier's pile-up, a one-in-25 chance. Among 301 bodies with radial non-gravitational terms, three are lighter than a rubble pile, each with a published explanation. No near-Earth object occupies the cells the delivery model leaves empty in a way dynamics forbids. The densest hyperbolic meteor group of the Global Meteor Network's 3.58 million orbits is matched by a 1.5 per cent speed bias, and its focal line holds no stream. NELIOTA's 284 lunar impact flashes are thermal and recur in place no more often than chance.

1 Introduction

The search for artifacts in the Solar System [1, 3] asks whether something made, a probe, a reflector, a habitat or a fragment, is among the bodies we already catalogue [4]. Surveys of the small bodies were not designed for that question, but they hold what it needs: radar echoes, reflectance spectra, lightcurves, phase curves, thermal fits, occultation chords, non-gravitational terms in the orbit solutions, and the orbits of meteoroids and of the near-Earth population. A made object can differ from a natural body in each: a flat, conducting or photovoltaic surface; a shape no collisional fragment has; a spin no rubble pile survives; an area too large for its mass; an orbit the natural sources do not reach. This paper puts each of these to the archive that measures it, with a null drawn from the same data, and states what each would have found.

1.1 Contributions

1. Tests of the surfaces of the small bodies: radar albedos, Gaia DR3 reflectance spectra against metal and photovoltaic templates with injected surfaces, glints in ALCDEF lightcurves, ATLAS phase curves and NEOWISE beaming parameters.
2. Tests of their shapes and spins: lightcurve amplitudes at zero phase, straight limbs in occultation profiles, the strength that the fastest spins need, and bodies that share a period, size and amplitude on unrelated orbits.

3. Tests of their orbits: area-to-mass ratios from non-gravitational terms, near-Earth objects where the delivery model has none, and coherent hyperbolic meteor streams.

4. A test of NELIOTA's lunar impact flashes for non-thermal colours and recurrence.

2 Surfaces

Radar. A flat metal plate facing the radar returns its echo in the opposite circular sense with a radar albedo near one, where a rough natural surface returns some power in the same sense and a stony one has an albedo near 0.1 [5]. The compilation of every published asteroid radar detection in the Small Bodies Node, 315 objects, holds nine albedos above 0.3: M- and X-class bodies, the radar-metallic 1986 DA [6], the E-types Nereus and 1998 WT24, and the Sq-type Bacchus at 0.33 ± 0.25. None combines an albedo near one with a same-sense to opposite-sense ratio near zero, the signature of a specular reflector.

Reflectance spectra. Gaia DR3 gives reflectance spectra of 60518 Solar System objects in 16 bands from 374 to 1034 nm [7], of which 58208 have twelve or more good bands from 418 to 990 nm. Each is fitted by the first four principal components of the sample, with and without the reflectance of a photovoltaic cell and of gold, copper, silver and aluminium from their optical constants [8] as an additive component. No spectrum takes a photovoltaic component, and the drops in χ^2 above 25 that the metal templates give fall on V, A and S types whose deep absorption bands the four components fit poorly. Injected into 3000 spectra, a photovoltaic surface is recovered in 41, 88 and 100 per cent of cases at 5, 10 and 20 per cent of the light at 550 nm; gold and copper in 19 and 9 per cent at 50 per cent of the light, and silver and aluminium, whose reflectance is nearly flat across the band, never.

Glints. A flat surface on a spinning body reflects the Sun toward us once a rotation, as a brief spike at a fixed rotational phase. In the ALCDEF photometry, 4723 sessions of 20 or more points at 15 min or less on bodies with an LCDB period [9] of quality 2 or better, spikes above the rotational lightcurve that recur at the body's period are compared with the same statistic for fades. No body's spikes recur at $p < 0.01$, where the fades, tested the same way, gather in ten groups. Injected spikes are recovered in 3, 29, 52 and 63 per cent of sessions at 0.05, 0.1, 0.2 and 0.5 mag.

Phase curves. A surface that shines with its own light, or that reflects as a smooth specular body does, brightens little with decreasing phase angle. The ATLAS phase curves in two bands [10] give 39239 fits of the H, G_1, G_2 function [11] spanning phase angles below 7° and above 25°, with a median slope of 0.0277 mag per degree from 5 to 30°. The 221 curves pinned at the flattest slope the function allows, 0.0179 mag per degree, have median residuals of 0.28 mag, rotation swamping the phase signal; the 107 of them with measured albedos have 0.03 to 0.74, none above the 0.8 that a flat curve on a bright surface would need to stand out.

Beaming. A body heated by more than the Sun shows a hotter dayside than a rough sunlit surface can, a NEATM [12] beaming parameter below about 0.6. Of the 74609 NEOWISE thermal fits with a free beaming parameter [13], of 68500 objects, eight fall below 0.6 at 2σ, each at a single epoch, and the lowest sits at the fit's floor of 0.3; the population's first percentile is 0.68, and no object stays below 0.6 at every epoch it was fitted.

3 Shapes and spins

Plates. A collisional fragment's lightcurve amplitude rarely exceeds 1.2 mag at low phase, while a plate of axis ratio 10 reaches 2.5 mag. In the LCDB's detail records, 31008 of quality 2 or 3 at phase angles up to 15°, with amplitudes reduced to zero phase at 0.03 mag per degree, five records

exceed 1.6 mag, 0.02 per cent against the 0.11 per cent of the quality-1 records, one exceeds 2.0 and none 2.5. All five are single TESS records [14] of slow rotators of 9 to 627 h, three marked ambiguous; no body exceeds a 4 : 1 axis ratio at low phase.

Straight limbs. A stellar occultation traces a body's limb chord by chord. The occultation bundle of the Small Bodies Node holds 395 events of fit quality 3 or 4 with five or more positive chords from distinct sites, of which 376 are measured once 19 whose chords disagree or whose fitted ellipse is far from the body's size are set aside. Each chord's ends are placed on the fundamental plane, and the longest run of four or more consecutive limb points that a line fits within the timing errors is measured against the diameter and against the fitted ellipse's points at the same angles and errors. The median run is 0.21 of the diameter and the 90th percentile 0.67. Twenty-five runs exceed the ellipse's null at $p < 0.01$, the longest on (3317) Paris in 2020 (0.97 of the diameter over six sites), (134) Sophrosyne in 1980 (0.86), (64) Angelina in 2023 (0.82), (39) Laetitia in 1998 (0.79) and (2) Pallas in 1983 (0.70 over six of 120 chords). Their shape models [15] carry the flat facets that an ellipse lacks, so the null that decides is the models projected at each event's geometry.

Spin strength. A gravitational aggregate of 2000 kg m^{-3} flies apart faster than about 2.3 h [16], and a faster spin needs cohesion that grows with size [17]. Of the 8854 asteroids with periods and diameters in the LCDB, no kilometre body spins faster than ten minutes or needs 100 kPa; the most demanding are 2019 MA2, 0.49 km across from its H and spinning in 2.62 min at quality 2, which needs 97 kPa, within a monolith's strength, and 2008 RA106, 3.7 km and 54 min at quality 2−, which needs 11 kPa. The secure fast rotators of the 31149 periods of quality 2 or 3 are the cohesive ones already published.

Replicated bodies. Objects made to one design would share a period, a size and a lightcurve amplitude on orbits that collisions do not relate. Among 4315 numbered asteroids with LCDB periods of quality 3 and the proper elements of the asteroid-family bundle [18], pairs that agree in period to 2×10^{-4}, in H to 0.1 mag and in amplitude to 0.05 mag while more than 300 m s^{-1} apart in the Zappalà metric [19] number 14, against 8.6 ± 2.8 with the periods shuffled within half-magnitude bins, and one group of three forms, against 0.04 ± 0.18, once in 25 shuffles: (3305) Ceadams, (5122) Mucha and (2019) van Albada, at 2.7290 to 2.7294 h, $H = 12.3$ to 12.4 and amplitudes 0.20 to 0.23, in three families. Their Gaia reflectance spectra are all S-type, with red slopes to 1.21 to 1.26 at 770 nm and the 1 μm band, and their period lies in the pile-up of fast rotators near the spin barrier, where periods crowd.

4 Orbits

Area-to-mass ratios. A radial non-gravitational term A_1 measures the push of sunlight or outgassing per unit mass, and for a body pushed by sunlight alone gives its area-to-mass ratio. Of the 301 objects with A_1 in JPL's database, three are lighter than a rubble pile under the most favourable assumptions: the active asteroid 362P/(2008 GO98), the interstellar object 1I/2017 U1 and the dark comet 2016 NJ33 [20], two of them with transverse or normal terms that mark activity.

Unreachable orbits. The delivery model of Granvik et al. [21] gives the density of near-Earth objects from each natural source over orbital elements, and leaves cells empty that no source reaches. The JPL database holds 37 near-Earth objects in those cells: 24 retrograde, all at eccentricities of 0.74 or more, the Halley-type comets and comet-like asteroids; none retrograde below 0.3; one near-circular orbit inside 0.75 au above 25°, 2025 LP, within the Atira population; and 12 above 60° with eccentricities below 0.25, which Kozai cycles [22] reach from lower inclinations.

Hyperbolic meteor streams. Meteoroids from another star, or debris shed by something travelling between stars, would arrive on one incoming asymptote at one speed. The Global Meteor

Network [23] holds 3.58 million orbits from 2018 to 2026, 40753 sporadic meteors hyperbolic at 3σ; each orbit gives its incoming asymptote and excess speed, and a meteor's neighbours within 1° and $3\,\mathrm{km\,s^{-1}}$ are counted against the density within 10°. Thirty-five meteors fall below $p = 10^{-4}$ in ten groups. Eight belong to IAU showers [24] whose speeds are overestimated (Perseids, Orionids, η Aquariids and the ξ Perseids). The largest, 16 meteors from 2022 to 2026 at solar longitudes 80 to 214° on one asymptote at ecliptic longitude 134.5° and latitude −1.6° at $12\,\mathrm{km\,s^{-1}}$, is matched by a 16-member group in one of three realisations of a 1.5 per cent speed overestimate shared by the elliptic sporadics, 74000 simulated meteors, and fifteen realisations thinned to the observed number at 1.0 to 2.0 per cent give 3 to 31 meteors below $p = 10^{-4}$ and groups of up to 14. SonotaCo's independent 353231 orbits [25] hold one meteor on that asymptote against 0.16 from the local density and about 1.6 that a real stream would give. A stream would also spread along the loci on which Earth's gravity focuses it, reaching ±52° of latitude, half of them south, as Earth crosses the asymptote near solar longitude 134.5°: the August 3 to 11 loci hold 529 meteors against a mean of 16 on 11 control asymptotes, but 510 of them lie north, where the Perseids (204 tagged) and the apex sporadics are, and 19 south, against 10 by chance. The group is the speed bias acting on the shower population, and no star within 100 pc shares its velocity on its line.

5 Lunar flashes

Impacts on the Moon's night side flash thermally for tens of milliseconds; a flash from a surface source, a lamp or a laser, would be hotter or cooler than the impact plume, longer, or recur at one place. NELIOTA's flash lists [26, 27] hold 284 flashes from 2017 to 2023, 188 of them validated and 192 with R and I photometry, located to 0.5°. Colour temperatures from $R - I$ through the Planck ratio, with Vega's colour as the zero point, have a median of 2935 K and a 5th to 95th percentile range of 2027 to 5020 K; none is hotter than 8000 or cooler than 1000 K at 3σ, and none lasts longer than 0.5 s, the longest 429 ms. Pairs within 1.5° number 21 against 23.9 ± 4.9 with the longitudes permuted among the flashes ($p = 0.75$); the closest, two validated flashes 0.10° apart on consecutive Perseid nights of 2018 August at (+26.7°, +60.2°), with temperatures of 4100 and 3000 K, is one that a tenth of a pair is expected to match.

6 Related work

Freitas [1] set out the search for extraterrestrial artifacts in the Solar System, Papagiannis [2] argued for exploring the asteroid belt for them, Haqq-Misra and Kopparapu [3] estimated how little of the Solar System has been searched at the sizes a probe would have, and Benford [4] proposed the co-orbital objects of the Earth as the likeliest places for one. The tests here rest on the archives' own studies: the radar survey of metal-rich asteroids [5], the Gaia reflectance spectra [7], the ATLAS phase curves [10], the spin barrier and its cohesive exceptions [16, 17], the dark comets [20], the debiased near-Earth population [21] and the meteor networks' orbit catalogues [23, 25], which this paper puts to the use those proposals describe.

7 Conclusion

No test finds an artifact among the small bodies. Every bright radar echo is a metallic or enstatite asteroid, no reflectance spectrum carries a photovoltaic surface, no lightcurve glints at its rotation period, no phase curve or thermal fit needs a source of light other than the Sun, no amplitude, limb or spin exceeds what natural shapes and strengths allow, the one coincidence of period, size and

Table 1: The thirteen tests, their samples and results.

Test	Sample	Result
Radar albedos	315 radar detections	albedos above 0.3 are M, X, E and one Sq type; no specular reflector
Reflectance spectra	58208 Gaia DR3 spectra	no photovoltaic or metal surface; 88 per cent recovery at 10 per cent of the light
Glints	4723 ALCDEF sessions	no spikes recurring at the rotation period at $p < 0.01$
Phase curves	39239 ATLAS curves	the flattest are rotation-dominated, albedos 0.03 to 0.74
Beaming	74609 NEOWISE fits	eight single-epoch fits below $\eta = 0.6$; none at every epoch
Plates	31008 LCDB records	none above 2.5 mag at zero phase
Straight limbs	395 occultations	longest runs on bodies with faceted shape models
Spin strength	8854 asteroids	no kilometre body under 10 min or above 100 kPa
Replicated bodies	4315 asteroids	one S-type triple at the spin barrier, one in 25
Area-to-mass	301 radial terms	three light bodies, each explained
Unreachable orbits	37 near-Earth objects	comets, an Atira and Kozai-reachable orbits
Hyperbolic streams	40753 hyperbolic meteors	groups matched by a 1.5 per cent speed bias; no focal line
Lunar flashes	284 NELIOTA flashes	thermal, 2000 to 5000 K; recurrence at chance

amplitude is the expected crowding at the spin barrier, every light body is explained, no near-Earth orbit lies beyond the natural sources' reach, the hyperbolic meteor groups are speed biases, and the lunar flashes are impacts.

The reach differs from test to test. The spectra exclude photovoltaic cover of a tenth of a body's light among bodies Gaia measured, down to about $G = 20$, and say nothing of silver or aluminium; the glint search reaches spikes of 0.1 mag in a third of the sessions; the radar sample is the 315 bodies radar has reached; the occultation test needs the shape-model null to separate a made facet from a natural one; and an object smaller or fainter than these surveys' limits, or one that keeps its distance from the inner Solar System, is not constrained.

Acknowledgements

This work uses the Planetary Data System Small Bodies Node (the asteroid radar compilation, the ALCDEF photometry, the Asteroid Lightcurve Database, the occultation bundle and the asteroid-family bundle), Gaia DR3 through CDS and ESA, the ATLAS phase curves through VizieR, the NEOWISE diameters and albedos from the NASA/IPAC Infrared Science Archive, JPL's Small-Body Database, the Global Meteor Network's trajectory summaries, SonotaCo's orbit catalogue, the IAU Meteor Data Center, the NELIOTA flash tables, the optical constants of refractiveindex.info, and the software NumPy, SciPy and astropy.

References

[1] R. A. Freitas, The search for extraterrestrial artifacts (SETA), *J. Br. Interplanet. Soc.* **36**, 501 (1983).

[2] M. D. Papagiannis, The importance of exploring the asteroid belt, *Acta Astronaut.* **10**, 709 (1983).

[3] J. Haqq-Misra and R. K. Kopparapu, On the likelihood of non-terrestrial artifacts in the Solar System, *Acta Astronaut.* **72**, 15 (2012).

[4] J. Benford, Looking for lurkers: co-orbiters as SETI observables, *Astron. J.* **158**, 150 (2019).

[5] M. K. Shepard, P. A. Taylor, M. C. Nolan, et al., A radar survey of M- and X-class asteroids. III. Insights into their composition, hydration state, and structure, *Icarus* **245**, 38 (2015).

[6] S. J. Ostro, D. B. Campbell, J. F. Chandler, et al., Asteroid 1986 DA: radar evidence for a metallic composition, *Science* **252**, 1399 (1991).

[7] Gaia Collaboration, L. Galluccio, M. Delbo, et al., Gaia Data Release 3: reflectance spectra of Solar System small bodies, *Astron. Astrophys.* **674**, A35 (2023).

[8] M. N. Polyanskiy, Refractiveindex.info database of optical constants, *Sci. Data* **11**, 94 (2024).

[9] B. D. Warner, A. W. Harris and P. Pravec, The asteroid lightcurve database, *Icarus* **202**, 134 (2009).

[10] M. Mahlke, B. Carry and L. Denneau, Asteroid phase curves from ATLAS dual-band photometry, *Icarus* **354**, 114094 (2021).

[11] K. Muinonen, I. N. Belskaya, A. Cellino, et al., A three-parameter magnitude phase function for asteroids, *Icarus* **209**, 542 (2010).

[12] A. W. Harris, A thermal model for near-Earth asteroids, *Icarus* **131**, 291 (1998).

[13] A. Mainzer, T. Grav, J. Masiero, et al., Thermal model calibration for minor planets observed with Wide-field Infrared Survey Explorer/NEOWISE, *Astrophys. J.* **736**, 100 (2011).

[14] A. Pál, R. Szakáts, Cs. Kiss, et al., Solar System objects observed with TESS: first data release, bright main-belt and Trojan asteroids from the southern survey, *Astrophys. J. Suppl.* **247**, 26 (2020).

[15] J. Ďurech, V. Sidorin and M. Kaasalainen, DAMIT: a database of asteroid models, *Astron. Astrophys.* **513**, A46 (2010).

[16] P. Pravec and A. W. Harris, Fast and slow rotation of asteroids, *Icarus* **148**, 12 (2000).

[17] K. A. Holsapple, Spin limits of Solar System bodies: from the small fast-rotators to 2003 EL61, *Icarus* **187**, 500 (2007).

[18] D. Nesvorný, M. Brož and V. Carruba, Identification and dynamical properties of asteroid families, in *Asteroids IV*, ed. P. Michel, F. E. DeMeo and W. F. Bottke (Univ. Arizona Press, 2015), p. 297.

[19] V. Zappalà, A. Cellino, P. Farinella and Z. Knežević, Asteroid families. I. Identification by hierarchical clustering and reliability assessment, *Astron. J.* **100**, 2030 (1990).

[20] D. Z. Seligman, D. Farnocchia, M. Micheli, et al., Dark comets? Unexpectedly large nongravitational accelerations on a sample of small asteroids, *Planet. Sci. J.* **4**, 35 (2023).

[21] M. Granvik, A. Morbidelli, R. Jedicke, et al., Debiased orbit and absolute-magnitude distributions for near-Earth objects, *Icarus* **312**, 181 (2018).

[22] Y. Kozai, Secular perturbations of asteroids with high inclination and eccentricity, *Astron. J.* **67**, 591 (1962).

[23] D. Vida, D. Šegon, P. S. Gural, et al., The Global Meteor Network: methodology and first results, *Mon. Not. R. Astron. Soc.* **506**, 5046 (2021).

[24] T. J. Jopek and Z. Kaňuchová, IAU Meteor Data Center: the shower database, a status report, *Planet. Space Sci.* **143**, 3 (2017).

[25] SonotaCo, A meteor shower catalog based on video observations in 2007–2008, *WGN* **37**, 55 (2009).

[26] A. Liakos, A. Z. Bonanos, E. M. Xilouris, et al., NELIOTA: methods, statistics, and results for meteoroids impacting the Moon, *Astron. Astrophys.* **633**, A112 (2020).

[27] A. Liakos, A. Z. Bonanos, E. M. Xilouris, et al., NELIOTA: new results and updated statistics after 6.5 years of lunar impact flashes monitoring, *Astron. Astrophys.* **687**, A14 (2024).